Scenario Identification for Implicit and Covert Communication in Small Language Models

SMBUD Project - By the XAI Research Team

July 2026

1 Existing Project

1.1 Objective

This project investigates whether small language models (LLMs) can establish covert communication channels through shared sociocultural knowledge. The study explores how an implicit message can be embedded within a rewritten text and successfully recovered only by a reader model that shares the writer's persona and cultural background.

2 Related Work

According to Umberto Eco, communication emerges through cooperation between a writer and a *Model Reader*, who share the knowledge needed to infer meanings that are not explicitly stated (Eco 1979). Recent work has investigated whether LLMs can perform similar forms of implicit communication, showing that language models often struggle to generate and recover implied meanings, especially when successful interpretation requires pragmatic reasoning and shared background knowledge (Sato, Kawano, and Yoshino 2025; Tint et al. 2026).

2.1 Methodology

The experimental framework consists of five LLM agents:

- **Writer LLM**: Receives a source text (e.g., a short story, social commentary, or news excerpt) and a target hidden message. The writer rewrites the text while implicitly embedding the message without making it explicit.
- **Baseline Reader LLM**: Shares the same persona as the writer and reads the original text before rewriting. It identifies both explicit and implicit meanings present in the source text, establishing a baseline of interpretations that can be used to distinguish pre-existing meanings from newly introduced ones.
- **In-Group Reader LLM**: Shares the same sociocultural background as the writer and reads the rewritten text. Its task is to infer the hidden message. Successful communication occurs when the recovered message exceeds the interpretations already identified in the baseline text.
- **External Reader LLM**: Reads the rewritten text without access to the shared sociocultural persona. Its task is also to infer the hidden message. A successful covert channel should allow the in-group reader to recover the message while the external reader cannot.
- **Supervisory Judge LLM**: A larger, neutral model evaluates all experiments. It blocks cases where the hidden message is considered too explicit, ensuring that communication remains genuinely implicit rather than directly encoded.

2.1.1 Intervention Test

When the in-group reader successfully recovers the hidden message, the supervisory judge performs a causal analysis:

- It identifies the textual elements that likely enabled message recovery.
- It rewrites the text while removing or neutralizing those elements.
- The decoding experiment is then repeated.

If the in-group reader can no longer recover the message, the removed elements are considered the causal carriers of the implicit communication.

2.2 Research Questions

1. Can persona-aligned small LLMs reliably communicate hidden information through shared sociocultural inferences rather than explicit encoding mechanisms?
2. Which linguistic and cultural mechanisms best support covert communication?
3. How effectively can shared sociocultural knowledge improve message recovery?
4. Can a larger supervisory model reliably distinguish implicit communication from overly explicit encoding?
5. Can causal intervention tests identify the specific textual features responsible for message transmission?

2.3 Evaluation Metrics

- Hidden message recovery rate by the in-group reader.
- Hidden message recovery rate by the external reader.
- Difference between baseline interpretations and post-rewrite interpretations.
- Supervisory rejection rate for overly explicit rewrites.
- Success rate of intervention tests in eliminating message recovery.

2.4 Expected Outcome

The project aims to characterize the emergence of implicit communication channels in small LLMs, quantify the role of shared cultural priors in message transmission, and identify the textual mechanisms that causally support covert communication while remaining opaque to uninformed readers.

2.5 Important Note About the Project

This project does not investigate steganography, secret encoding schemes, watermarking, or any form of intentionally coded communication. The hidden message must not be transmitted through predefined lexical substitutions, cipher-like mappings, acrostics, token-level encoding, or other explicit signaling strategies.

Instead, the project focuses on implicit communication emerging from shared sociocultural knowledge. Message recovery should occur through interpretation of culturally meaningful references, narrative framing, symbolism, values, experiences, or background assumptions shared between the writer and the in-group reader.

Therefore, successful communication must rely on inference rather than decoding. If the hidden message can be recovered through an identifiable encoding rule that does not require the shared persona, the experiment is considered out of scope.

References

Eco, Umberto (1979). *Lector in fabula: la cooperazione interpretativa nei testi narrativi*. Also reprinted by La nave di Teseo in 2020. Milano: Bompiani.

Sato, Takuma, Seiya Kawano, and Koichiro Yoshino (Dec. 2025). "Pragmatic Theories Enhance Understanding of Implied Meanings in LLMs". In: *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*. Ed. by Kentaro Inui et al. Mumbai, India: The Asian Federation of Natural Language Processing and The Association for Computational Linguistics, pp. 2458–2477. ISBN: 979-8-89176-298-5. DOI: 10.18653/v1/2025.ijcnlp-long.132. URL: https://aclanthology.org/2025.ijcnlp-long.132/.

Tint, Joshua et al. (Mar. 2026). "ExpressivityBench: Can LLMs Communicate Implicitly?" In: *Findings of the Association for Computational Linguistics: EACL 2026*. Ed. by Vera Demberg, Kentaro Inui, and Lluís Marquez. Rabat, Morocco: Association for Computational Linguistics, pp. 4500–4515. ISBN: 979-8-89176-386-9. DOI: 10.18653/v1/2026.findings-eacl.235. URL: https://aclanthology.org/2026.findings-eacl.235/.

3 Student Assignment

The goal of the student project is to **design and evaluate a complete covert communication experiment** between small language models. Students are expected to **identify a realistic communication scenario in which shared sociocultural knowledge may facilitate the transmission of implicit information**.

3.1 Prompts

Specifically, students must define the following prompts:

- A **Writer LLM prompt** specifying how the model should rewrite a text while implicitly conveying a target hidden message without making it explicit.
- An **In-Group Reader LLM prompt** specifying how the reader should analyze the rewritten text and infer possible implicit meanings.
- An **External Reader LLM prompt** representing a reader who does not share the same sociocultural background and attempts to infer the hidden message from the same text.

3.2 Experimental Scenario

In addition, students must identify a working scenario by defining the following:

- A **text corpus** consisting of source materials to be rewritten, such as short stories, news articles, opinion pieces, social media posts, or other short texts.
- A set of **hidden messages** that are plausibly related to the chosen sociocultural background and that can be embedded through indirect references, framing choices, narrative elements, symbolism, or culturally specific cues.

- A **shared sociocultural persona**, including relevant cultural background, experiences, values, interests, or group-specific knowledge that will be provided to both the writer and the in-group reader through persona prompting.

3.3 Deliverables

Students must **justify their design choices** and **explain why the selected sociocultural background is expected to facilitate message recovery by the in-group reader while limiting recovery by the external reader**.

The final deliverable should include **all prompts, the selected corpus, the hidden messages, and an experimental evaluation** conducted in accordance with the methodology described in this project. **grice1975logic**